

Aerial-Ground Person Re-Identification Leaderboards

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Abstract. This document presents leaderboard tables summarizing reported experimental results on five Aerial-Ground Re-Identification (AG-ReID) benchmarks. All results are collected from **formally published papers**. Therefore, Results reported only in arXiv preprints are therefore not included.

Keywords: Aerial-ground person re-identification · Leaderboards

Overview

The leaderboard tables are designed to provide a **concise, visually clear and publication-ready** comparison format. **Each table reserves the final row for users to insert their own experimental results**, and the best result under each evaluation metric is highlighted in **bold**. For readability, the reported methods in each table are divided into three groups: representative CNN-based methods, representative ViT-based methods, and methods specifically designed for AG-ReID.

Usage

The file `ag_reid_leaderboards.tex` in the current directory is the L^AT_EX source of `ag_reid_leaderboards.pdf`. Users may directly copy the corresponding tables from this source file for use in their own manuscripts.

Table 1: Performance comparison on AG-ReID. Best results are in **bold**.

Method	A→G		G→A	
	mAP	R1	mAP	R1
BoT [12]	55.47	70.01	58.83	71.20
OsNet [31]	58.32	72.59	60.99	74.22
SBS [3]	59.77	73.54	62.27	73.70
VV [6]	67.23	77.22	69.83	79.73
ViT [2]	72.61	81.47	73.39	82.85
TransReID [4]	73.23	81.33	75.71	84.10
CLIP-ReID [9]	75.68	82.59	78.51	85.68
Explain [14]	72.38	81.28	73.35	82.64
VDT [29]	74.44	82.91	78.57	86.59
DTST [20]	74.51	83.48	76.05	84.10
GSAIalign [8]	75.01	83.75	77.73	84.10
VIF [5]	75.22	83.75	79.19	87.32
SD-ReID [21]	75.40	85.16	77.02	85.57
SAS-VPReID [25]	75.98	84.24	77.28	86.17
SeCap [19]	76.16	84.03	78.34	87.01
LATex [27]	77.67	85.26	81.15	89.40
SVPR-ReID [30]	77.85	85.34	80.55	87.32
HiHR [24]	79.21	86.06	81.28	87.94
CVAF [13]	79.45	87.23	82.59	90.12
GLPSG [32]	81.50	87.70	84.30	90.60
Yours				

Table 2: Performance comparison on AG-ReID v2. Best results are in **bold**.

Method	A→C		C→A		A→W		W→A	
	mAP	R1	mAP	R1	mAP	R1	mAP	R1
Swin [11]	57.66	68.76	57.70	68.80	56.15	68.49	53.90	64.40
HRNet-18 [18]	65.07	75.21	66.16	76.25	66.17	76.26	66.17	76.25
SwinV2 [10]	66.09	76.44	62.14	77.11	69.09	80.08	65.61	74.53
MGN(R50) [17]	70.17	82.09	72.41	84.21	78.66	88.14	73.73	84.06
BoT(R50) [12]	71.49	80.73	69.67	79.46	75.98	86.06	72.41	82.69
SBS(R50) [3]	72.04	81.96	73.89	84.10	78.94	88.14	75.01	84.66
PCL-CLIP [7]	72.20	79.80	72.40	81.12	77.70	87.14	73.89	84.19
BoT(ViT) [12]	77.03	85.40	75.90	84.65	80.48	89.77	75.90	84.65
ViT [2]	77.03	85.40	79.30	89.77	80.48	89.77	77.42	88.24
CLIP-ReID [9]	79.79	85.36	79.08	85.64	84.23	89.14	79.55	86.50
FusionReID [22]	80.70	86.70	80.00	87.90	84.20	89.70	80.90	86.50
TransReID [4]	81.40	88.00	80.10	87.60	84.50	90.40	81.10	87.70
SAS-VPReID [25]	76.39	83.40	78.47	85.26	82.10	89.45	78.47	85.73
Explain [14]	79.00	87.70	78.24	87.35	83.14	93.67	79.08	87.73
VDT [29]	79.13	86.46	78.12	86.14	82.21	90.00	78.52	85.26
SD-ReID [21]	80.61	87.04	79.24	86.74	84.06	90.86	80.12	86.79
V2E [15]	80.72	88.77	79.51	87.86	84.85	93.62	80.11	88.61
SeCap [19]	80.84	88.12	78.99	88.24	84.01	91.44	80.15	87.56
GSAIalign [8]	81.38	87.86	81.05	88.02	83.98	90.63	80.90	87.31
CVAF [13]	83.35	88.77	83.17	88.40	87.81	92.49	85.08	90.85
LATex [27]	83.50	89.13	82.85	89.01	86.35	91.35	83.30	89.32
ViSA [28]	83.61	89.43	82.32	88.57	85.99	91.63	83.10	89.23
HiHR [24]	84.04	88.84	83.21	88.57	87.57	91.40	84.02	89.50
Yours								

Table 3: Performance comparison on CARGO. Best results are in **bold**.

Method	ALL		A→G		A→A		G→G	
	mAP	R1	mAP	R1	mAP	R1	mAP	R1
BAU [1]	38.40	45.20	36.70	40.40	42.60	50.00	51.20	61.60
VV [6]	38.44	45.83	29.00	31.25	49.73	67.50	62.99	72.31
SBS [3]	43.09	50.32	29.00	31.25	49.73	67.50	62.99	72.31
PCB [16]	44.50	51.00	30.40	34.40	44.60	55.00	67.60	74.10
BoT [12]	46.49	54.81	32.56	36.25	49.79	65.00	66.47	77.68
MGN [17]	49.08	54.81	33.47	31.87	52.96	65.00	71.05	83.93
AGW [26]	53.44	60.26	40.90	43.57	56.48	67.50	71.66	81.25
ViT [2]	53.54	61.54	40.11	43.13	64.47	80.00	71.34	82.14
PCL-CLIP [7]	60.93	67.31	51.43	54.37	60.75	70.00	76.00	84.82
CLIP-ReID [9]	64.25	68.27	53.83	55.62	65.42	75.00	80.80	84.82
VDT [29]	55.20	64.10	42.76	48.12	66.83	82.50	71.59	82.14
DTST [20]	55.73	64.42	43.49	50.53	63.31	80.00	72.40	78.57
VIF [5]	57.46	72.76	44.55	51.25	66.98	82.50	74.19	83.93
SD-ReID [21]	57.47	65.06	46.44	53.12	67.70	82.50	74.08	81.25
GSAIalign [8]	57.95	65.06	61.55	64.89	65.55	80.00	73.86	83.04
SAS-VPReID [25]	58.48	64.42	57.36	59.51	63.41	70.00	76.90	83.93
SeCap [19]	60.19	68.59	58.94	68.43	68.08	80.00	75.42	86.61
ViSA [28]	65.46	70.51	69.00	71.28	67.78	82.50	83.90	88.39
LATex [27]	67.09	76.96	58.88	66.87	69.06	80.00	79.90	90.18
HiHR [24]	67.52	74.36	70.53	75.53	69.60	75.00	80.93	87.50
Yours								

Table 4: Performance comparison on LAGPeR. Best results are in **bold**.

Method	A→G		G→A		G→AG	
	mAP	R1	mAP	R1	mAP	R1
CLIP-ReID [9]	17.60	24.40	20.10	21.30	10.20	12.30
ViT [2]	27.25	38.67	30.69	32.04	15.31	18.88
TransReID [4]	28.80	38.80	32.10	33.00	18.80	22.90
MIP [23]	29.30	39.30	32.60	33.90	17.30	21.00
Explain [14]	28.89	40.48	31.91	32.96	18.91	22.33
VDT [29]	28.97	40.15	31.91	33.55	16.45	19.50
SD-ReID [21]	29.72	40.15	32.86	34.47	18.88	22.85
SeCap [19]	30.37	41.79	33.42	35.26	19.24	24.39
ViSA [28]	30.42	41.37	33.59	35.06	20.31	25.61
SAS-VPReID [25]	30.76	43.83	33.40	35.42	23.02	31.12
HiHR [24]	33.80	46.95	36.74	39.30	23.99	31.39
Yours						

Table 5: Performance comparison on CP2108. Best results are in **bold**.

Method	ALL		A→G		G→A	
	mAP	R1	mAP	R1	mAP	R1

SBS [3]	17.29	27.96	23.05	36.06	23.83	39.14
MGN [17]	23.93	32.45	34.02	52.27	34.26	52.58
BoT [12]	28.35	36.27	38.60	57.76	37.72	55.64
ViT [2]	37.92	49.36	42.03	61.14	44.58	61.97
TransReID [4]	38.18	50.32	45.62	62.03	45.00	62.28
CLIP-ReID [9]	41.12	57.90	45.35	62.16	45.32	62.97
VDT [29]	33.10	46.88	33.64	42.76	31.74	40.59
SeCap [19]	38.74	53.90	46.95	63.55	46.19	63.39
SVPR-ReID [30]	42.40	59.23	47.72	64.25	47.04	63.76
Yours						

References

- Cho, Y., Kim, J., Kim, W.J., Jung, J., Yoon, S.e.: Generalizable person re-identification via balancing alignment and uniformity. *Advances in Neural Information Processing Systems* **37**, 47069–47093 (2024)
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., et al.: An image is worth 16x16 words: Transformers for image recognition at scale. In: *International Conference on Learning Representations* (2020)
- He, L., Liao, X., Liu, W., Liu, X., Cheng, P., Mei, T.: Fastreid: A pytorch toolbox for general instance re-identification. In: *Proceedings of the 31st ACM International Conference on Multimedia*. pp. 9664–9667 (2023)
- He, S., Luo, H., Wang, P., Wang, F., Li, H., Jiang, W.: Transreid: Transformer-based object re-identification. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 15013–15022 (2021)
- Khalid, W., Liu, B., Li, X., Waqas, M., Afgan, M.S.: Bridging the sky and ground: Towards view-invariant feature learning for aerial-ground person re-identification. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 9749–9758 (2025)
- Kumar, R., Weill, E., Aghdasi, F., Sriram, P.: A strong and efficient baseline for vehicle re-identification using deep triplet embedding. *Journal of Artificial Intelligence and Soft Computing Research* **10**(1), 27–45 (2020)
- Li, J., Gong, X.: Prototypical contrastive learning-based clip fine-tuning for object re-identification. *arXiv preprint arXiv:2310.17218* (2023)
- Li, Q., Li, J., Zhang, Y., Tan, L., Chen, J., Ji, J.: Gsalgn: Geometric and semantic alignment network for aerial-ground person re-identification. *Advances in Neural Information Processing Systems* **38**, 93785–93805 (2026)
- Li, S., Sun, L., Li, Q.: Clip-reid: exploiting vision-language model for image re-identification without concrete text labels. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 37, pp. 1405–1413 (2023)
- Liu, Z., Hu, H., Lin, Y., Yao, S., Xie, Z., Wei, Y., Ning, J., Cao, Y., Zhang, Z., Dong, L., et al.: Swin transformer v2: Scaling up capacity and resolution. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 12009–12019 (2022)
- Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B.: Swin transformer: Hierarchical vision transformer using shifted windows. In: *Proceedings of the IEEE/CVF International Conference on Computer Vision*. pp. 10012–10022 (2021)
- Luo, H., Gu, Y., Liao, X., Lai, S., Jiang, W.: Bag of tricks and a strong baseline for deep person re-identification. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops*. pp. 1487–1495 (2019)
- Mao, D., Teng, S., Lyu, X.: Cvaf: A clip-based view-consistent alignment framework for aerial-ground person re-identification. *ACM Transactions on Multimedia Computing, Communications and Applications* **22**(3), 1–19 (2026)
- Nguyen, H., Nguyen, K., Sridharan, S., Fookes, C.: Aerial-ground person re-id. In: *2023 IEEE International Conference on Multimedia and Expo*. pp. 2585–2590 (2023)
- Nguyen, H., Nguyen, K., Sridharan, S., Fookes, C.: Ag-reid. v2: Bridging aerial and ground views for person re-identification. *IEEE Transactions on Information Forensics and Security* **19**, 2896–2908 (2024)
- Sun, Y., Zheng, L., Li, Y., Yang, Y., Tian, Q., Wang, S.: Learning part-based convolutional features for person re-identification. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **43**(3), 902–917 (2019)
- Wang, G., Yuan, Y., Chen, X., Li, J., Zhou, X.: Learning discriminative features with multiple granularities for person re-identification. In: *Proceedings of the 26th ACM International Conference on Multimedia*. pp. 274–282 (2018)
- Wang, J., Sun, K., Cheng, T., Jiang, B., Deng, C., Zhao, Y., Liu, D., Mu, Y., Tan, M., Wang, X., et al.: Deep high-resolution representation learning for visual recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **43**(10), 3349–3364 (2020)
- Wang, S., Wang, Y., Wu, R., Jiao, B., Wang, W., Wang, P.: Secap: self-calibrating and adaptive prompts for cross-view person re-identification in aerial-ground networks. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 22119–22128 (2025)
- Wang, Y., Pishgar, M.: Dynamic token selection for aerial-ground person re-identification. *arXiv preprint arXiv:2412.00433* (2024)
- Wang, Y., Hu, X., Wang, L., Zhang, P., Lu, H.: Sd-reid: view-aware stable diffusion for aerial-ground person re-identification. *arXiv preprint arXiv:2504.09549* (2025)
- Wang, Y., Zhang, P., Liu, X., Tu, Z., Lu, H.: Unity is strength: Unifying convolutional and transformer features for better person re-identification. *IEEE Transactions on Intelligent Transportation Systems* pp. 3713–3723 (2025)
- Wu, R., Jiao, B., Wang, W., Liu, M., Wang, P.: Enhancing visible-infrared person re-identification with modality- and instance-aware visual prompt learning. In: *Proceedings of the ACM International Conference on Multimedia*. pp. 579–588 (2024)
- Yang, Q.: Hierarchical hyperbolic representation for aerial-ground person re-identification (2026), *arXiv preprint, arXiv ID pending*
- Yang, Q., Zhang, P., Wang, Y., Gong, Z.: Sas-vpreid: A scale-adaptive framework with shape priors for video-based person re-identification at extreme far distances. In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV) Workshops*. pp. 1599–1608 (March 2026)
- Ye, M., Shen, J., Lin, G., Xiang, T., Shao, L., Hoi, S.C.: Deep learning for person re-identification: A survey and outlook. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**(6), 2872–2893 (2021)
- Zhang, P., Hu, X., Wang, Y., Lu, H.: Latex: Leveraging attribute-based text knowledge for aerial-ground person re-identification. *arXiv preprint arXiv:2503.23722* (2025)
- Zhang, Q., Cai, Z., Zhao, P., Wu, J., Wu, C., Chen, H., Lai, J.: View-aware semantic alignment for aerial-ground person re-identification. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 4383–4392 (2026)
- Zhang, Q., Wang, L., Patel, V.M., Xie, X., Lai, J.: View-decoupled transformer for person re-identification under aerial-ground camera network. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. pp. 22000–22009 (2024)
- Zheng, A., Xie, H., Wan, X., Wang, Z., Li, S., Tang, J., Luo, B.: Semantic-driven visual progressive refinement for aerial-ground person reid: A challenging large-scale benchmark. In: *Proceedings of the AAAI Conference on Artificial Intelligence*. vol. 40, pp. 13360–13368 (2026). <https://doi.org/10.1609/aaai.v40i16.38339>
- Zhou, K., Yang, Y., Cavallaro, A., Xiang, T.: Learning generalisable omni-scale representations for person re-identification. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**(9), 5056–5069 (2021)
- Zhu, R., Chen, H., Xie, X., Lai, J.: Global-local prompts-driven semantic guidance for aerial-ground person re-identification. In: *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*. pp. 99–112. Springer (2025)